

# Count-On Detectives

## Answer Key

Kindergarten–Grade 1

### Quick Answers

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|---------------------------|----------------------------|
| 1. 8                      | 4. the correct sum = 10, — |
| 2. the correct sum = 8, — | 5. the correct sum = 9, —  |
| 3. the correct sum = 9, — | 6. the sum = 9, —          |
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### What is verified

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1 of 6 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 2, 3, 4, 5, 6. The worked solution states what a correct response must contain.

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### Curriculum

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**Level:** Kindergarten–Grade 1

K.OA.A – problems 1, 2, 3, 4, 5, 6

*Difficulty 1–4 of 5 across 11 tagged checks.*

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### Common wrong answers

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- 2. *If they got 7: counted the start number 5 as the first count, landing one short*
  - 3. *If they got 8: started the count at 6 instead of counting on after it, landing one short*
  - 4. *If they got 9: said 8 as the first hop instead of starting the hops after 8*
  - 5. *If they got 8: counted the start number as a hop, one short of the true sum*
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**1.** Warm-up. Count on the right way: for  $6 + 2$ , say “6”, then hop two times. What do you land on?

**Solution:** Say 6 (no hop yet!), then hop: 7, 8. Two hops for the 2 being added, landing on 8. So  $6 + 2 = 8$ . 8

**2.** Jen added  $5 + 3$ . She counted: “5, 6, 7” — and wrote  $5 + 3 = 7$ . Write the correct sum, and tell why Jen’s count went wrong.

**Solution:** Jen said the start number 5 as her first count, so she only hopped 2 times for the 3 being added. Count on correctly: say 5, then hop 6, 7, 8 — three hops. So  $5 + 3 = 8$ . 8

*Tell why (instructor judges):* a good answer says Jen counted the start number 5 as a hop, so she landed one short; the hops should be 6, 7, 8.

**3.** Kim added  $6 + 3$ . She counted: “6, 7, 8” — and wrote  $6 + 3 = 8$ . Find Kim’s mistake. Write the correct sum, and show the count she should have said.

**Solution:** Kim’s first count was the start number 6, so only two of her counts were real hops. Say 6 first, then hop 7, 8, 9. So  $6 + 3 = 9$ . 9

*Find the mistake (instructor judges):* the student points at the “6” in Kim’s count as the error and writes the correct count 7, 8, 9.

**4.** Raj added  $8 + 2$ . He counted: “8, 9” — and wrote  $8 + 2 = 9$ . Cross out the wrong sum, write the correct one, and tell how Raj should count.

**Solution:** Raj used 8 as his first hop. Say 8 without counting it, then hop 9, 10 — two hops for the 2 being added. So  $8 + 2 = 10$ . 10

*Tell how (instructor judges):* a good answer says Raj should say 8 first, then count exactly two hops: 9, 10.

**5.** Ava says  $4 + 5 = 8$ . Check her sum by counting on from 5. Is Ava right? Write yes or no, then write the correct sum.

**Solution:** Check by counting on from the bigger number 5: say 5, then hop 6, 7, 8, 9 — four hops for the 4. Ava is *not* right: she landed one short at 8. The true sum is  $4 + 5 = 9$ . 9

*Is she right (instructor judges):* the student writes “no”, shows the count 6, 7, 8, 9, and fixes the sum to 9.

**6.** Challenge! Leo says: “I never make the count-on mistake, because I say the start number first, and then I count my hops on my fingers.” Add  $3 + 6$  Leo’s way, starting from the bigger number. Then tell how saying the start number first stops the mistake.

**Solution:** Start big: say 6 first. Then count three hops on three fingers: 7, 8, 9. So  $3 + 6 = 9$ . 9

*Tell how (instructor judges):* a good answer says that saying the start number before any hops keeps it from being counted as a hop, so the number of hops matches the number being added — three fingers for the 3.